

MICHAŁ HORECKI

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Technical Skills

Programming: C++, CUDA, Python (PyTorch, NumPy), Dart/Flutter

Tools: Git, CMake

Spoken: English (C1), Polish (native), German (basic)

Projects

GradCraft: Deep Learning Framework from Scratch | C++, CUDA | [GitHub](#)

May 2026 – September 2026

- Trained MALLMOC, a 90M-parameter GPT, on a single RTX 3090 with 1.5B tokens of python-edu (about 20 tokens/param). It converged and writes simple Python snippets. Training throughput reached 45–50% of PyTorch eager mode (no FlashAttention), measured on a 180M-parameter model.
- Built a lazily evaluated autograd engine with zero-copy tensor views and aliasing, plus a PyTorch-style frontend (Module, Parameter, AdamW). No external AI dependencies.
- Wrote memory pools for CPU and CUDA, so training loops don't allocate memory.
- Fused chains of primitive ops into single CUDA kernels. One RMSNorm kernel replaces nine dispatches, and the fused RMSNorm and softmax cross-entropy ran up to about 100x faster than the unfused versions.

Tiled GEMM + Tiled Attention | CUDA

September 2026

- Wrote a tiled GEMM kernel and a tiled attention kernel in CUDA (shared memory, register tiling, no vectorized loads).
- Benchmarked on an RTX 3090. GEMM is about 10x slower than cuBLAS. Attention is about 24x slower than PyTorch's FlashAttention backend and about 4x slower than its math backend, which writes the full $T \times T$ matrix.

Lyme Disease Detection Pipeline | TensorFlow, Computer Vision

October 2025 – January 2026

- Built a 3-stage image pipeline as a paid project. A filter rejects photos without people (trained on 16k positive and 16k negative images), a checker rejects photos not viable for assessment (10k and 10k), and a specialist model detects lyme.
- The filter and checker together reach an accuracy of about 99.8%. The specialist, trained on 596 lyme and 13,300 non-lyme photos, achieved 88.7% recall at 85% specificity on a held-out set (47 TP, 6 FN, 407 TN, 73 FP).
- Preprocessed images with white balance and brightness/contrast, and chose the best model using PR AUC and recall, because the data was heavily imbalanced. I also tuned the decision threshold for each model to maximize correctness.

Speech Denoiser (LSTM) | Python, NumPy | [GitHub](#)

January 2026

- Built a speech denoiser in NumPy with no ML library, including a BiLSTM with handwritten backprop through time (BPTT) and my own STFT/ISTFT.
- The network predicts only the difference between the noisy and clean spectrograms, not the whole spectrogram.
Audio samples are on [GitHub](#).

Experience

Mathwise (EdTech Startup)

June 2024 – February 2026

Software Developer

FlutterFlow, Flutter

- Main developer and de facto tech lead for the first year. Decided features with the founder and guided the other three developers.
- Built the app's frontend and application logic in FlutterFlow, then Flutter. The app reached about 8,000 users, with about 500 paying simultaneously at its peak.
- Built an AI feature using the Gemini Vision API. It extracts bounding boxes from users' solutions and checks each step for mistakes.

Education

Warsaw University of Technology (EiT)

Computer Science (Informatyka), BSc Eng

Warsaw, Poland

October 2026 – February 2030

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High School

Warsaw, Poland

September 2022 – May 2026